

1. Administrative & Study Identification

Full Study Title: Patient characteristics, treatment patterns and outcomes of patients with severe asthma treated with tezepelumab in the real world; a retrospective, observational medical chart review conducted in UK severe asthma centres **Short Title/Acronym:** TRAILBLAZE **Protocol Number:** D5180R00027 **NCT Number:** NCT06455462 **Sponsor Name:** AstraZeneca **Investigational Product:** Tezepelumab (Tezspire) **Phase of Development:** Not Applicable **Trial Status:** Recruiting **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate and describe the real-world clinical outcomes, specifically measuring the annualized exacerbation rate, of severe asthma patients treated with tezepelumab. **Secondary Objectives:** To characterize patient demographics, treatment patterns, and broader clinical outcomes in the 52 weeks following the initiation of tezepelumab therapy.

2.2 Background & Rationale To evaluate the real-world utilization and effectiveness of tezepelumab. This study provides necessary observational data on patients participating in the tezepelumab patient access programme (TPAP) and routine care to understand tezepelumab's impact on severe asthma (SA) control outside of controlled interventional clinical trials.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational (Retrospective chart review) **Control Type:** N/A (Single-arm descriptive) **Blinding:** N/A (Open-label / Observational) **Randomization:** N/A (Non-randomized)

3.2 Planned Enrollment & Duration **Target \$N\$:** 350 enrolled patients **Number of Sites:** Up to eleven NHS trusts in the United Kingdom **Estimated Study Start:** 22-Apr-2024 **Estimated Primary Completion:** 25-Nov-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients aged ≥ 18 years at index. 2. Patients who received their first dose of tezepelumab between 1st January 2023 and 19th July 2023 (TPAP cohort) or between 20th July 2023 and 31st March 2024 (post-TPAP cohort).

4.2 Exclusion Criteria 1. Patients who received any biologic drug for the treatment of asthma in a clinical trial at any time during the 52 weeks prior to the index date.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Details: Tezepelumab (Trade Name: Tezspire | ATC Code: R03DX11) **Dosage/Frequency:** Tezepelumab administration as prescribed per routine clinical care and the TPAP guidelines. **Route of Administration:** Subcutaneous (standard for tezepelumab). **Duration of Treatment:** Follow-up observation outcomes period of 52 weeks post-index date.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care background therapies for severe asthma. **Prohibited:** Concurrent participation in a biologic clinical trial during the 52-week baseline period prior to the index date.

6. Study Timeline & Visit Schedule

(Note: As a retrospective chart review, "visits" represent data abstraction periods rather than prospective patient encounters)

Visit ID	Window	Key Activities/Procedures		:---	:---
:---		Pre-index Period	Any time prior to tezepelumab initiation	Historical clinical data abstraction prior to initiation of therapy	
		Baseline Period	52 Weeks Prior to Index Date	Abstraction of historical clinical data, exacerbation history, and prior treatments	
		Index Date	Day 0	First dose of tezepelumab administered	
		Outcomes Period	52 Weeks Post-Index Date	Retrospective abstraction of exacerbations, clinical outcomes, and treatment patterns via eCRFs	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualised exacerbation rate during the 52-week post-index outcomes period. **Measurement Method:** Medical chart review and eCRF abstraction of clinical health records.

7.2 Secondary & Exploratory Endpoints 1. Exacerbation rate during the baseline. 2. Change from baseline in annual exacerbation rate. 3. Exacerbation rate by season.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Retrospective extraction of documented AEs from patient medical records via eCRFs. **Serious**

Adverse Events (SAEs): Extracted via historical chart review as recorded by the treating healthcare professional. **Data Monitoring Committee (DMC):** N/A (Retrospective observational chart review).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible patients with an index date enrolled from the selected NHS trusts. **Sample Size Justification:** Approximately 350 patients to provide descriptive real-world evidence and evaluate clinical parameters comprehensively. **Handling of Missing Data:** Standard missing data handling protocols and exclusion principles for retrospective medical chart data abstraction.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local patient privacy regulations. **Monitoring Plan:** Retrospective data abstraction monitored for quality and completeness via electronic case report forms (eCRFs). **Audit Trail:** Maintained within the eCRF database with source data verification against NHS trust health records.

11. Study Identifiers & Governance

Primary ID: D5180R00027 **Registry Number:** NCT06455462 **Posting ID:** 444840344075762326 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** TRAILBLAZE **Official Regulatory Title:** Patient characteristics, treatment patterns and outcomes of patients with severe asthma treated with tezepelumab in the real world; a retrospective, observational medical chart review conducted in UK severe asthma centres

12. Clinical Framework (Severe Asthma Specific)

Primary Condition: Severe Asthma (SA) (MeSH: D060235 | SNOMED CT: 370221004) **Diagnosis Threshold:** Clinical diagnosis qualifying for the Tezepelumab Patient Access Programme (TPAP) or standard care **Medical Methodology:** Real-world observational chart review **Optimization Target:** Exacerbation reduction and overall asthma control

13. Study Procedures & Methodology

Management Pathway: UK NHS standard clinical pathway for severe asthma **Education Protocol (HCP):** N/A (Retrospective chart review) **Education Protocol (Patient):** N/A (Retrospective chart review)

Quality Audit Mechanism: Data abstraction and validation via electronic case report forms (eCRFs)

14. Observation & Follow-Up Schedule

Total Duration: Patients will be followed up from their index date until the first of the following events (whichever is first): they reach 52 weeks post-index, they switch to a different biologic treatment, they die, or they are otherwise lost to follow-up. **Onsite Visit Cadence:** N/A (Retrospective data collection) **Remote Monitoring Frequency:** N/A **Checklist Review Interval:** N/A

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					